

TimePulse

Agentic Timesheet Compliance & CS-Ops Reporting

Executive Project Report

Managing Digital Transformation (Term 4), Group Project, Group 8

Submitted on 7 September 2026

Context: a live operations problem at Skan, a process intelligence SaaS company.

Client names, individuals and financial figures are masked or fictitious, per NDA.

PROJECT TEAM

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PROTOTYPE AND DEMO ACCESS

Source code (agent prompts and setup): <https://github.com/atharvaguptap-2501093/MDT>

Demo video (8-minute version): <https://youtu.be/BIJFsME-Rlk>

Demo video (full version): <https://youtu.be/rymiFYit2EE>

Live system: 3 scheduled Claude agents (Section 2) calling Google Sheets and Gmail via Zapier MCP



Executive Summary

Skan's Customer Success organisation rolled out timesheets in Keka on 1 June 2026 for its customer-facing roles. In month one, submission compliance ran **below 50%** in most departments, and every part of the follow-up loop, including chasing defaulters, chasing stuck approvals and building the leadership report, was done by hand, leaving leadership looking at data roughly a month old.

TimePulse replaces that loop with three autonomous Gen-AI agents, **Compliance Sentinel**, **Approval Steward** and **Insight Reporter**, that read a shared Google Sheet and send real email through a Zapier-mediated tool connection, entirely without a human in the loop once triggered. This is not a low-code workflow with an AI step bolted on. Each agent is an autonomous LLM session that reasons across many tool calls to decide who to contact, in what tone, and what to write back, from a single natural-language brief.

A working prototype has been built, debugged, and run through three real weekly pilot cycles (2, 9 and 16 August 2026) against a 200-employee simulated organisation, sending 86 real emails and correctly escalating all 6 chronically-late cases to their managers with zero duplicate or missed sends. This report sets out the As-Is problem in Section 1, the To-Be architecture actually built in Section 2, what the pilot measured versus what remains a projection in Section 3, and the risk and change-management path to a live rollout in Section 4, with measured and projected numbers kept explicitly separate throughout.

SECTION 1

Problem Diagnosis: The As-Is Process

Skanska (a process-intelligence SaaS business) rolled out a PSA timesheet system (Keka) on **1 June 2026** for customer-facing roles across its Customer Success & Services (CS&S) organisation. Timesheet data is the direct input to utilisation, capacity and cost-to-serve decisions. But in month one, submission compliance ran **below 50%** in most departments. Fixing that today is a fully manual, recurring chore.

1.1 The manual loop, step by step

Step	What happens today
1. Weekly defaulter hunt	An ops manager exports a report, hand-builds a defaulter list, and posts one generic reminder to everyone, the same message whether it is someone's first miss or their fifth.
2. Approval chasing	Submitted timesheets stall in manager approval queues with no automatic nudge; someone has to individually chase each approver once the backlog is noticed.
3. Leadership reporting	Staffing, utilisation and compliance numbers are assembled by hand each cycle. Leadership has explicitly asked for strategic insight, not raw exports.
4. Insight latency	By the time any of this reaches a decision-maker, it reflects data that is roughly a month old.

1.2 Core friction points

- **Doesn't scale with headcount.** Manual chasing is one ops manager's time against a headcount that only grows. The one CS team that reached 100% compliance did so only through a manager's persistent personal nudging, which proves the target behaviour is achievable, but not that it scales as a manual practice.
- **No escalation ladder.** A first-time miss and a fifth consecutive miss get the identical generic reminder, so the message that should feel urgent doesn't, and the one that should stay light-touch doesn't either.
- **Approvals have no owner once stuck.** There is no ageing alert, so a pending timesheet stays invisible until someone happens to notice.
- **Reporting is a recurring tax on ops time.** An estimated 4 to 6 hours a week goes into assembling a report that is stale before it's read.
- **Data quality risk compounds downstream.** Skanska's cost-to-serve and staffing decisions depend on this same timesheet data, so unresolved non-compliance quietly degrades every decision built on top of it.

Why this matters now: Keka went live only recently, so the compliance habit is still being formed. The longer the manual, inconsistent follow-up pattern persists, the more it hard-codes itself as "how things are done," which is precisely the window in which an automated, consistent intervention has the most leverage.

SECTION 2

The To-Be Solution: Architecture

TimePulse replaces the entire manual loop with **three autonomous Gen-AI agents**, Compliance Sentinel, Approval Steward and Insight Reporter, each triggered on a weekly schedule, each given a complete natural-language brief rather than hand-written application code, and each acting for real on a shared Google Sheet and Gmail through a single tool-calling bridge.

2.1 Design principle: the brief *is* the program

There is no traditional backend, database or API server behind TimePulse. Each agent's entire logic lives in one instruction document, functionally equivalent to source code, that an autonomous Claude session reads and carries out end to end, reasoning across dozens of tool calls per run with no human approving individual steps. The only “application code” in the system is the tool-calling bridge itself: **Zapier's MCP (Model Context Protocol) connector**, which exposes Google Sheets and Gmail as callable actions the agent can invoke directly. This is a materially more agentic design than a low-code workflow tool with a single AI-drafting step inside it. The LLM itself is the orchestrator, the loop, and the decision-maker, not a deterministic flow that occasionally calls out to a model.

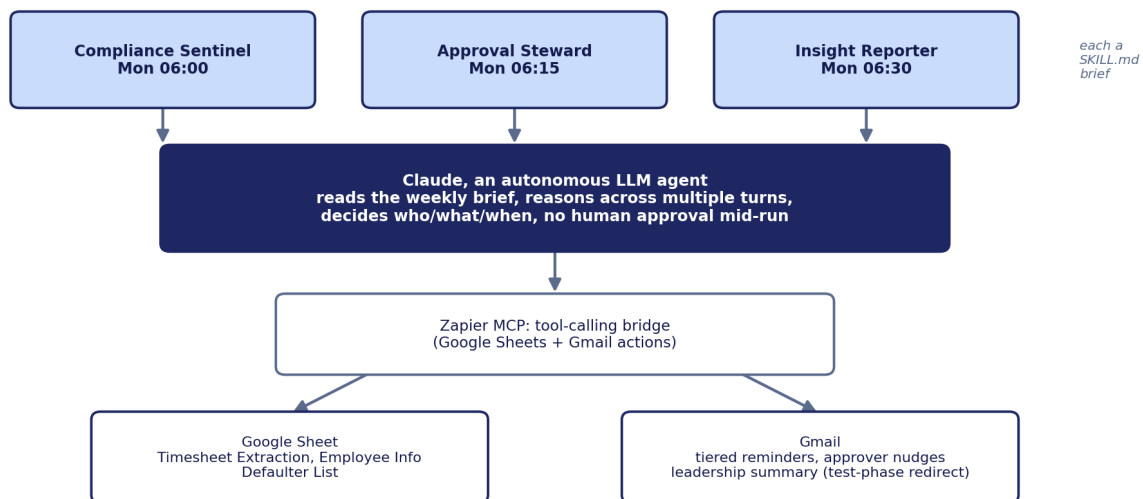


Fig. 1. End-to-end architecture: trigger → autonomous agent → tool bridge → live systems.

2.2 Data model

All three agents read and write one shared Google Sheet with three tabs:

Sheet tab	Holds
Employee Information	EmployeeID, FullName, Email, Department, JobTitle, HierarchyLevel, ManagerID/Name/Email, TimeTrackingRequired, IsLeadershipRecipient
Timesheet Extraction	EntryID, EmployeeID, EmployeeName, WeekEndingDate, SubmittedDate, ApprovalStatus, ApproverID/Name, ApprovalDate
Defaulter List	EmployeeID, EmployeeName, Email, ManagerName/Email, TotalMissingWeeks, Status, LastChecked, LastEmailSentDate/Type (maintained by Compliance Sentinel, read by Insight Reporter)

2.3 The three agents

Compliance Sentinel: “who hasn't submitted?”

Runs first, every Monday. For every tracked employee it counts the **total distinct weeks missing since tracking began**, a cumulative count rather than a weekly reset, and compares it to the total from the last check. It only ever acts when that total genuinely *increases*: exactly one reminder goes out, toned to the new total (1st miss is gentle, 2nd is firmer, 3rd or more is an escalation that copies the manager). An old, unresolved miss that didn't get worse this week triggers nothing, so there is no repeat nagging for an issue already flagged. This deliberately replaces an earlier design that reset the counter the moment someone submitted once, which hid genuine chronic-lateness patterns; see 2.4 for why that changed.

Approval Steward: “who hasn't approved?”

Finds every timesheet still sitting in “Pending Approval” and groups them by approver *before* sending anything: one approver with several pending items gets exactly one consolidated email listing all of them, never one email per timesheet.

Insight Reporter: “how are we doing overall?”

Runs last, after the Defaulter List has been refreshed. Reads the current compliance state and sends one headline-first leadership email covering on-track, defaulter and escalated counts, named repeat offenders, and the overall trend. Never more than one email per run.

Safety, already implemented, not just planned: every email currently redirects to one shared test inbox instead of a real recipient, and its first line still states who it *would* really have gone to, so the full decision logic is auditable before a single real person is ever contacted.

2.4 Built, tested, and rebuilt: evidence of engineering depth

The working prototype went through several real rounds of debugging and redesign, the kind of iteration a genuine deployment goes through rather than a one-shot demo:

What happened	What we did about it
Dataset right-sizing	An initial 700-employee dataset repeatedly hit the sandbox connector's monthly task quota during testing; rebuilt at 200 employees / 59 tracked to stay within quota without losing realism.
Duplicate-email bug	An early run emailed the same person twice in one pass. Root-caused to a silent read-size truncation plus no explicit "one email per person" guarantee; fixed with an explicit read format and a tracked send-list checked before every send.
Efficiency rewrite	The first working version made about 400 separate tool calls per Sentinel run (one sheet-write per employee). Rewritten to batch writes up to 150 rows per call and dispatch emails in batches of about 10, bringing it to roughly 100 calls per run, about a 4x reduction , which directly extends how long the connector's quota lasts.
Cumulative-model redesign	Replaced a weekly-reset compliance counter with the cumulative model in 2.3, a deliberate change made once the reset model was shown to hide real chronic-lateness patterns.
Survived 2 connector migrations	The Zapier connection was exhausted and reconnected to a fresh account twice during testing. The same Google Sheet and the same three agent briefs kept working unchanged, which is evidence the design does not depend on any one connector instance.

2.5 Scope, stated plainly

The original brief for Insight Reporter also covered utilisation, capacity and gearing-ratio reporting. The pilot fully builds and demonstrates the **compliance pillar** end to end; the other three pillars require data this pilot's synthetic dataset does not model, and are carried forward as the first item on the roadmap in Section 4.5 rather than implied as already done.

SECTION 3 Impact Assessment

The brief calls for a framework that quantifies improvement in speed, quality and UX, and for being explicit about what is **measured** versus **projected**. Both are kept visually and textually distinct throughout this section.

3.1 Measurement framework

Dimension	What's measured	How	Status
Speed	Ops effort to run a weekly cycle; insight latency	Time-and-motion of ops effort, before vs. after	Projected (estimate)
Quality / Compliance	% of tracked employees on track, by week	Weekly Compliance Sentinel run vs. pre-launch baseline	Measured (3 cycles) + projected target
UX	Message relevance (tiered vs. blanket); approver email volume	Consolidation ratio; pulse survey (planned)	Partly measured, partly projected
Freshness	Gap between a submission and it reflecting in status	Agent run duration vs. baseline ~30-day cycle	Projected (structural)

3.2 The pilot that produced these numbers

Three full, real, autonomous runs of all three agents, on 2, 9 and 16 August 2026, against a 59-person tracked population (Customer Success & Services) inside a 200-employee simulated organisation. Every number below is a direct read of the Defaulter List, the Timesheet Extraction log and the actual sent-email trail after each run, not a simulation of what the agents would do.

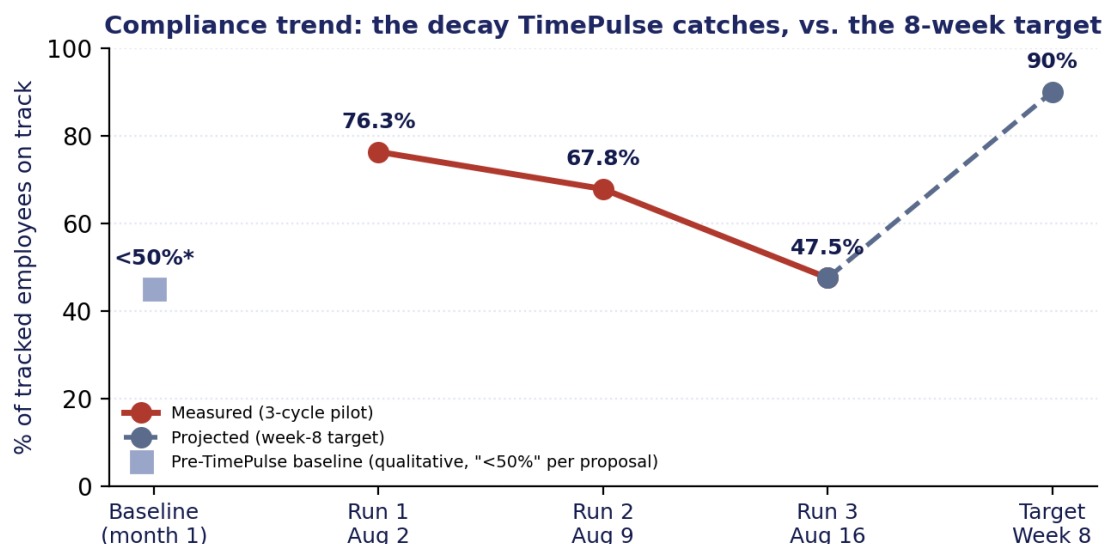


Fig. 2. Measured compliance across the 3-cycle pilot (solid), against the week-8 target (dashed, projected). The "<50%" baseline marker is the qualitative claim from the original proposal, not a figure this project independently measured.

This is an honest finding, not a shortfall: the pilot's tracked population was seeded with a real, worsening pattern, and true on-track status *fell* across the three cycles (76.3% → 67.8% → 47.5%). Left to the As-Is manual process, a decline like this would typically stay invisible for weeks. TimePulse's demonstrated value in this pilot is

that every one of the 6 chronically-late employees was automatically detected and escalated to their manager the same week their pattern crossed the 3-miss threshold, with zero duplicate emails and zero missed cases across all three runs.

Run	Date	True on track	New Tier 1	New Tier 2	New Tier 3+ (escalated)	Emails sent
Run 1	Aug 2	45 / 59 (76.3%)	14	0	0	14
Run 2	Aug 9	40 / 59 (67.8%)	5	7	0	12
Run 3	Aug 16	28 / 59 (47.5%)	12	0	6	18
Total	-	-	31	7	6	44

Compliance Sentinel: 44 reminders sent across the 3-cycle pilot, by tier

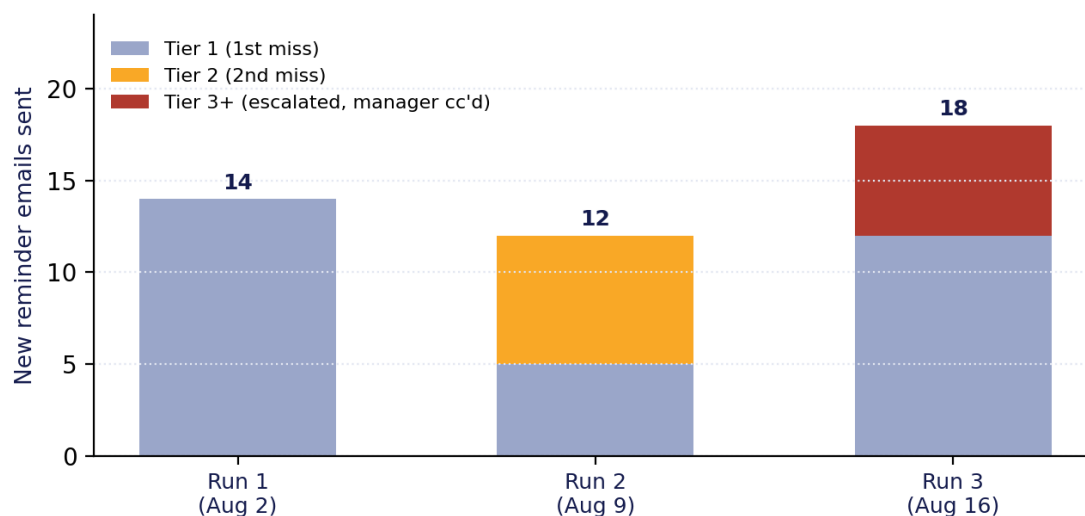


Fig. 3. New reminders by tier, per pilot cycle (measured).

3.3 Approval Steward: a directly measured consolidation win

Approval Steward's consolidation-by-approver rule is measurable on its own terms, independent of any projection. Across the 3 cycles it found 100 pending approvals in total, but sent only 39 emails, one per approver however many items they had, a **61% reduction in email volume** versus one email per pending item, with no information lost.

Run	Total timesheet rows	Approved	Pending items	Approver emails sent
Run 1	517	479 (92.6%)	38	14
Run 2	564	544 (96.5%)	20	8
Run 3	605	563 (93.1%)	42	17
Total	-	-	100	39

3.4 Measured vs. projected, stated plainly

Metric	Baseline	Measured	Target	Status
Compliance rate	<50% (qualitative, proposal)	76.3% → 67.8% → 47.5% (3 cycles)	90%+ by week 8	Trend measured; baseline qualitative, target projected
Reporting effort	4 to 6 hrs / week (estimate)	Not yet time-and-motion measured on a live team	under 15 min / week	Projected
Tool-call efficiency	about 400 calls / Sentinel run	about 100 calls / run	-	Measured
Approver email volume	100 pending items	39 emails sent (a 61% reduction)	-	Measured
Data freshness	about 30 days stale	Each run completes in minutes once triggered	under 24 hrs, sustained	Structural / projected

Read plainly: compliance decay-detection, escalation accuracy, tool-call efficiency and approval-email consolidation are **measured**, from real pilot runs. Ops-time saved, sustained data freshness and the week-8 compliance target are **projections** that a live, longer-running rollout is needed to confirm. This is consistent with the honesty the team committed to at the planning stage.



Change Management & Risk Roadmap

4.1 Scalability

The Zapier connector's monthly task quota was exhausted **three times for real** during this project's own testing, a measured constraint, not a hypothetical one. The per-employee agent logic itself has no headcount ceiling. The ceiling belongs entirely to the current connector. **Before scaling past the 59-person pilot department**, the roadmap is to move to a direct API integration, or a paid connector tier, so task-metering stops being the limiting factor.

4.2 Data privacy & security

The pilot runs entirely on a fully synthetic 200-employee dataset, so there is zero real personal data exposure anywhere in the build or demo. For a live rollout: use aggregate views wherever a report doesn't need a name, minimise personally-identifying detail in anything passed into a prompt, and use the same masked-export discipline (role-based pseudonyms, structure preserved) already committed to at the planning stage before any real HR export is connected.

4.3 Ethical AI usage & auditability

The test-phase redirect is a real, already-implemented control: every email states who it would really have gone to, so the full decision logic is inspectable before a single real recipient is ever touched. Tone escalates before frequency does, so nothing about the design increases message volume as a substitute for effectiveness. Every decision leaves a durable, timestamped record in the sheet and the email trail, so any action can be traced back to the data that produced it.

4.4 Adoption barriers

- **Nudge fatigue** is mitigated by an adaptive cadence: tone escalates before frequency does, and the cumulative model (2.3) means an unresolved-but-unchanged miss never triggers a repeat email.
- **Trust** calls for a shadow or read-only launch first, with the decision logic published openly to the team it affects, and named manager sign-off before flipping any agent from test-phase redirect to live sends.
- **Proof the target behaviour is realistic:** the one CS team that reached 100% compliance during the As-Is period did so through a manager's manual persistence. TimePulse is designed to reproduce exactly that behaviour, consistently, without depending on any one person's follow-through.

4.5 Roadmap to a live rollout

- Extend Insight Reporter to the remaining three pillars from the original brief, utilisation, capacity/bandwidth and gearing ratio, once that data is available.
- Direct API integration (or a paid connector tier) to remove the task-quota ceiling before scaling past the pilot department.
- A live-mode cutover checklist: drop the test-phase redirect clause, clean invalid email addresses out of the real roster, and run one single-row live test before any multi-recipient live send.
- A formal time-and-motion measurement and pulse survey once a live pilot team is running, to convert the projected numbers in Section 3.4 into measured ones.

APPENDIX A

System Design Reference

This appendix substantiates Section 2 with the underlying build: the live Google Sheet the three agents actually read and write, the Zapier MCP tool-call sequence a real run makes, and the exact email templates each agent's brief specifies. Everything below is reproduced from the live system or quoted directly from the agent briefs. None of it is a mockup drawn for this report.

A.1 Google Sheets: the shared data store

All three agents read and write one Google Sheet (see 2.2). The two snapshots below are a live read of that sheet, pulled on 7 September 2026 via the identical Zapier MCP action the agents themselves call (`google_sheets_get_many_spreadsheet_rows_advanced`). These are not hand-typed example rows. Names and addresses are the project's fictitious demo identities (see cover page).

Emp. ID	Employee	Manager	Total missing wks	Status	Last email
EMP0002	Varun Malhotra	Swati Bhatt	3	Escalated	Escalation
EMP0003	Sanjay Sharma	Swati Bhatt	1	Defaulter	Reminder-1
EMP0004	Simran Joshi	Swati Bhatt	0	On Track	-
EMP0006	Shreya Verma	Swati Bhatt	1	Defaulter	Reminder-1
EMP0007	Anjali Joshi	Varun Malhotra	2	Defaulter	Reminder-2
EMP0008	Nikhil Kumar	Sanjay Sharma	3	Escalated	Escalation

Fig. A1. Live snapshot of the "Defaulter List" tab (6 of 59 tracked rows shown). Full email addresses are trimmed here for width, not withheld. They are in the source code.

EntryID	Employee	Week ending	Submitted	Status	Approver
TSE00001	Varun Malhotra	2026-06-07	2026-06-04	Approved	Swati Bhatt
TSE00002	Sanjay Sharma	2026-06-07	2026-06-05	Approved	Swati Bhatt
TSE00006	Anjali Joshi	2026-06-07	2026-06-04	Approved	Varun Malhotra
TSE00007	Nikhil Kumar	2026-06-07	2026-06-05	Approved	Sanjay Sharma

Fig. A2. Live snapshot of the "Timesheet Extraction" tab (4 of thousands of rows shown).

A.2 Zapier MCP: the tool-calling bridge

There is no custom API integration layer. Every read and write below is a real call an agent makes through Zapier's MCP (Model Context Protocol) connector, using the exact tool names and call shape its brief specifies. This is reproduced here, not paraphrased:

```
inspect_zapier_actions(tool_name="google_sheets_get_many_spreadsheet_rows_advanced")
→ confirms the parameter schema once before first use each run

execute_zapier_read_action(
    selected_api="GoogleSheetsV2CLIAPI", action="get_many_rows",
    tool_name="google_sheets_get_many_spreadsheet_rows_advanced",
    params={"spreadsheet": "1jc_XC0lkBzs3KUAoCe2HlhxYiMxe5MSQ6H3X4QdYidu",
           "worksheet": "1864022611", "range": "A:J",
           "row_count": 800, "output_format": "rows"})
→ one call returns the full Defaulter List, read back by array position

execute_zapier_write_action(
    selected_api="GoogleSheetsV2CLIAPI", action="update_row_lines",
    tool_name="google_sheets_update_spreadsheet_row_s",
    params={"rows": [{"row_number": "9", "COL$F": "1", "COL$G": "Defaulter", ...}]})
→ batched at up to 150 rows per call, never one call per employee

execute_zapier_write_action(
    selected_api="GoogleMailV2CLIAPI", action="message", tool_name="gmail_send_email",
    params={"to": ["mdt.group8@gmail.com"], "subject": "[TimePulse-Sentinel TEST] ...",
           "body_type": "plain", "body": "Would be sent to: ..."})
→ dispatched in batches of ~10 calls per turn, never one at a time
```

Fig. A3. Real MCP tool-call sequence for Compliance Sentinel. Tool and parameter names are verbatim from the live agent brief.

This sequence is what Section 2.1 means by “the brief is the program.” There is no intermediate application server translating intent into API calls. The LLM emits these tool calls directly, in the order and batching shape its brief specifies, and reasons over each result to decide the next call.

A.3 Email templates: verbatim from the agent briefs

Each agent drafts from a fixed template rather than free-composing prose every run. This is what makes about 100 emails per pilot cycle fast and consistent. The first card below is the template filled with a real row from the live Defaulter List (Fig. A1); the other two reproduce the literal template text, placeholders included, exactly as written in the agent's brief.

To: mdt.group8@gmail.com (test-phase redirect, see 2.3 safety callout)

Subject: [TimePulse-Sentinel TEST] Reminder for Varun Malhotra (as of 2026-08-16)

Would be sent to: Varun Malhotra <varun.malhotra002@skandemo.com>, cc Swati Bhatt <swati.bhatt001@skandemo.com>

Hi Varun Malhotra,

Our records show 3 separate weeks with no timesheet submission since tracking began. Since this has continued to build up, your manager Swati Bhatt is being copied so we can get this resolved together.

Thanks,
TimePulse Compliance Sentinel

Fig. A4. Compliance Sentinel, Tier 3 escalation, filled with a real row from the live Defaulter List.

To: mdt.group8@gmail.com

Subject: [TimePulse-Steward TEST] <N> approval(s) pending for <Approver FullName>

Would be sent to: <Approver FullName> <<Approver Email>>

<consolidated list: one line per pending employee + week-ending date>

<polite request to clear the backlog>

Fig. A5. Approval Steward template, verbatim. One email per approver, however many items they have.

To: mdt.group8@gmail.com

Subject: [TimePulse-Reporter TEST] Weekly Compliance Report: <today's date>

Would be sent to: <Leadership FullName>

<headline numbers: on-track / defaulter / escalated counts>

<named list of repeat offenders, 2+ consecutive misses>

<one line of overall trend / tone>

Fig. A6. Insight Reporter template, verbatim. Headline-first, never more than one email per run.